

Tax Evasion and Productivity

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Recap

Last meeting: What is an interesting question to wrap up the JMP?

- ▶ Counterfactual question: Optimal tax rate

Today:

- ▶ SOC of firm's optimization problem
- ▶ Functional forms for detection probability and tax evasion cost
- ▶ Estimation strategy

Introduction

- ▶ What's the question?
- ▶ Why should you care?

Contribution

- ▶ What do I do?
- ▶ How do I do it?

Overview of Results

- ▶ What do I find?
- ▶ Why is it important?

Colombia

Model

The Model

$$\begin{aligned} \max_{K_{it}, L_{it}, M_{it}, e_{it}} \quad & (1 - q(e_{it})) \left[P_t \mathbb{E}_\varepsilon[Y_{it}] - \rho_t M_{it} - r_t K_{it} - w_t L_{it} \right. \\ & \left. - \tau (P_t \mathbb{E}_\varepsilon[Y_{it}] - \rho_t (M_{it} + e_{it})) \right] \\ & + q(e_{it}) \left[(1 - \tau) (P_t \mathbb{E}_\varepsilon[Y_{it}] - \rho_t M_{it}) - r_t K_{it} - w_t L_{it} \right. \\ & \left. - \phi \rho_t \tau e_{it} \right] - \kappa_{it}(e_{it}, \omega_{it}) \\ \text{s.t.} \quad & Y_{it} = K_{it}^{\alpha_k} L_{it}^{\alpha_L} M_{it}^\beta \exp(\omega_{it} + \varepsilon_{it}) \end{aligned} \tag{1}$$

Important Assumptions

- ▶ Following seminal work of Allingham and Sandmo (1972), tax authority audits firms at random and always detects tax evasion
 - ▶ The probability of detection, $q(e_{it})$, is a function of the level of tax evasion, e_{it} . It might depend on other exogenous variables such as the monitoring technology of the tax authority, for example.
 - ▶ Impose assumption from Empirical model which assumes that choosing M does not enter the probability of detection or the cost of evasion. In other words, choosing M is independent of choosing e.

Important Assumptions (Cont.)

The cost of evasion, $\kappa_{it}(e_{it}, \omega_{it})$: any resources of the firm allocated to the tax evasion activity that could have been employed in a productive activity (e.g. manager's time spent on tax evasion instead of managing the firm).

- ▶ Unlike the penalty of tax evasion, ϕ , it does not depend on whether you are detected or not
- ▶ Increasing in the level of tax evasion, e_{it}
- ▶ Decreasing in firm's productivity, ω_{it} , but spikes at the top of the productivity distribution.

Important Assumptions - discussion on κ

- ▶ More productive firms have more resources to allocate to tax evasion, but since huge firms have access to more sophisticated legal tax avoidance strategies, the cost of illegal tax evasion is higher for the most productive firms.
- ▶ It could also be the case that the probability of detection is flat with respect to productivity except for huge (top productive) firms which could have a probability close to 1.
- ▶ Empirically, without data on cost of evasion or auditing information, q and κ are not separately identified. During estimation, we assume functional forms that allow us to reproduce the empirical patterns of tax evasion across the productivity distribution.

Model (Cont.)

The firms' problem (Equation 1) can be rewritten as follows,

$$\begin{aligned} \max_{K_{it}, L_{it}, M_{it}, e_{it}} \quad & \mathbb{E}[\pi_{it} | \Gamma] = (1 - \tau) (P_t \mathbb{E}[Y_{it}] - \rho_t M_{it}) - r_t K_{it} \\ & - w_t L_{it} + [1 - q(e_{it})] (\rho_t \tau e_{it}) - q(e_{it}) \phi \rho_t \tau e_{it} \\ & - \kappa_{it}(e_{it}, \omega_{it}) \\ \text{s.t.} \quad & Y_{it} = K_{it}^{\alpha_K} L_{it}^{\alpha_L} M_{it}^{\beta} \exp(\omega_{it} + \varepsilon_{it}) \end{aligned} \tag{2}$$

First Order Condition (FOC)

An interior solution leads to following first-order conditions,

$$\beta K_{it}^{\alpha_K} L_{it}^{\alpha_L} M_{it}^{-(1-\beta)} \exp(\omega_{it}) = \frac{\rho_t}{P_t} \quad (3)$$

$$\tau \rho_t (1 - (1 + \phi) (q(e_{it}) + q'(e_{it})e_{it})) = \frac{\partial \kappa_{it}(e_{it}, \omega_{it})}{\partial e_{it}} \quad (4)$$

(ε_{ijt} is measurement error, so $\mathcal{E} \equiv E[\exp(\varepsilon_{ijt})] = 1$)

Tax Evasion

To better understand the incentives to overreport inputs, we can rewrite Equation 4 in terms of the evasion elasticity of the probability of detection, $\xi_{q,e} = \frac{\partial q(e_{it})}{\partial e_{it}} \frac{e_{it}}{q(e_{it})}$, as follows,

$$\tau \rho_t \left(1 - (1 + \phi) (1 + \xi_{q,e}) q(e_{it}) \right) = \frac{\partial \kappa_{it}(e_{it}, \omega_{it})}{\partial e_{it}} \quad (5)$$

Corner Solution

Note that some firms will optimally choose not to evade, $e_{it} = 0$.

For these firms, the marginal cost of tax evasion is greater than the marginal benefit for any non-zero level of tax evasion.

This implies that for these firms, Equation 4 becomes,

$$\tau \rho_t \leq \frac{\partial \kappa_{it}(0, \omega_{it})}{\partial e_{it}} \quad (6)$$

Second Order Condition (SOC)

To guarantee a unique solution, the hessian matrix of the firm's problem must be negative definite.

$$\nabla^2 \mathbb{E}[\pi_{it} | \Gamma] = \begin{bmatrix} \frac{\partial^2 \mathbb{E}[\pi_{it} | \Gamma]}{\partial e_{it}^2} & 0 & 0 & 0 \\ 0 & \frac{\partial^2 \mathbb{E}[\pi_{it} | \Gamma]}{\partial M_{it}^2} & \frac{\partial^2 \mathbb{E}[\pi_{it} | \Gamma]}{\partial M_{it} \partial L_{it}} & \frac{\partial^2 \mathbb{E}[\pi_{it} | \Gamma]}{\partial M_{it} \partial K_{it}} \\ 0 & \frac{\partial^2 \mathbb{E}[\pi_{it} | \Gamma]}{\partial L_{it} \partial M_{it}} & \frac{\partial^2 \mathbb{E}[\pi_{it} | \Gamma]}{\partial L_{it}^2} & \frac{\partial^2 \mathbb{E}[\pi_{it} | \Gamma]}{\partial L_{it} \partial K_{it}} \\ 0 & \frac{\partial^2 \mathbb{E}[\pi_{it} | \Gamma]}{\partial K_{it} \partial M_{it}} & \frac{\partial^2 \mathbb{E}[\pi_{it} | \Gamma]}{\partial K_{it} \partial L_{it}} & \frac{\partial^2 \mathbb{E}[\pi_{it} | \Gamma]}{\partial K_{it}^2} \end{bmatrix} \quad (7)$$

- ▶ For a negative definite hessian matrix, the leading principal minors must alternate signs,
 - ▶ odd order minors must be negative and even-order minors positive.
- ▶ In our case, since
 - ▶ 1) the production function is strictly concave (CD) and
 - ▶ 2) choosing the level of tax evasion does not depend on the production function inputs; second cross derivatives are zero,
 - ▶ we would only have to check the first leading principal minor —the second derivative with respect to e_{it} — is negative,

$$\frac{\partial^2 \mathbb{E}[\pi_{it} | \Gamma]}{\partial e_{it}^2} = -\tau \rho_t (1 + \phi) (2q'(e_{it}) + e_{it} q''(e_{it})) - \frac{\partial^2 \kappa_{it}(e_{it}, \omega_{it})}{\partial e_{it}^2} < 0 \quad (8)$$

For strict concavity, we need only one of the following two conditions to be strictly positive,

$$\frac{\partial^2 \kappa_{it}(e_{it}, \omega_{it})}{\partial e_{it}^2} \geq 0 \quad (9)$$

$$\tau \rho_t (1 + \phi) (2q'(e_{it}) + e_{it} q''(e_{it})) \geq 0 \quad (10)$$

The second condition can be simplified to

$$-\frac{q''(e_{it})}{q'(e_{it})}e_{it} < 2 \quad (11)$$

since economic theory implies that $q'(e_{it}) > 0$ and $e_{it} > 0$, this condition imposes a **restriction on the curvature** of the detection function, $q(e_{it})$

- ▶ **Linear** detection **always** satisfies the SOC, since $q''(e_{it}) = 0$.
- ▶ **Convex** also satisfies the SOC **always**, since $q''(e_{it}) > 0$.
- ▶ **Concave** works if not too concave, i.e. **inequality holds**.

What does this inequality imply in practice?

With $\phi = 0$, logs of the interior FOC $\tau\rho_t(1 - q - q'e) = \partial\kappa/\partial e$ define

- ▶ the **log marginal benefit** $h(e) \equiv \ln(\tau\rho_t) + \ln(1 - q(e) - q'(e)e)$,
- ▶ with FOC $h(e) = \ln(\partial\kappa/\partial e)$.

	$q(e)$	SOC	$h(e) - \ln(\tau\rho_t)$	FOC ceiling
Linear	$\min\{\lambda e, 1\}$	$0 < 2$	$\ln(1 - 2\lambda e)$	$\lambda e < \frac{1}{2}$
Concave	$1 - e^{-\lambda e}$	$\lambda e < 2$	$-\lambda e + \ln(1 - \lambda e)$	$\lambda e < 1$
Convex	$\min\{(\lambda e)^k, 1\}$ $k > 1$	$1 - k < 2$	$\ln(1 - (1+k)(\lambda e)^k)$	$\lambda e < (1+k)^{-1/k}$

- ▶ All satisfy the SOC on their interior (exp's $\lambda e < 2 \iff \lambda e < 1$);
- ▶ All impose a restriction on λe .

A detection technology: the audit threshold

- ▶ The firm chooses overreporting e .
- ▶ The tax authority draws a threshold $T \sim U[0, 1/\lambda]$ and audits the firm if $e \geq T$.
 - ▶ An audit always uncovers evasion (Allingham and Sandmo 1972).
 - ▶ T is random to the firm (audit selection, third-party cross-checks, which inspector).

$$q(e) = \Pr(T \leq e) = \min\{\lambda e, 1\}.$$

- ▶ **Threshold scale:** $1/\lambda$ is the largest overreporting any audit draw tolerates; the *average* audit-triggering threshold is $1/2\lambda$.
- ▶ **Evasion bound:** the FOC stops firms at $\lambda e < \frac{1}{2}$
 - ▶ the most aggressive evader overreports below the *average* audit-triggering threshold,
 - ▶ at threshold, $q(1/2\lambda) = 50\%$ (a coin flip)

If $q(e)$ is concave, $T \sim \text{Exp}(\lambda)$ and $q(e) = 1 - e^{-\lambda e}$, mean threshold $1/\lambda$, detection $\approx 63\%$ at $e = 1/\lambda$

How likely is the most aggressive evader caught?

At the FOC ceiling, the largest evasion drives the marginal benefit to zero,
 $q(b) + q'(b)b = 1$.

	b	$q(b)$	$\xi_{q,e} _{e=b}$	$E[T]$
Linear	$\frac{1}{2\lambda}$	50%	1 (unit elastic)	$\frac{1}{2\lambda}$
Exponential	$\frac{1}{\lambda}$	63%	$\frac{\lambda b}{e^{\lambda b} - 1} < 1$ (inelastic)	$\frac{1}{\lambda}$
Power	$\frac{1}{\lambda(1+k)^{1/k}}$	$\frac{1}{1+k} < 50\%$	$k > 1$ (elastic)	$\frac{k}{\lambda(k+1)}$

- ▶ $\xi_{q,e} = q'(e)e/q(e)$ is the evasion elasticity of the detection probability
- ▶ $\frac{1}{e-1} = 0.58$

Discussion

- ▶ Power (convex) form produces the lowest average audit-triggering threshold,
 - ▶ Exponential (concave) the largest.
 - ▶ Linear is intuitive: at max evasion, the tax authority flips a coin to audit/caught the firm
- ▶ An increase in 1% in the level of evasion increases the probability of being caught, by
 - ▶ 1% if linear (constant)
 - ▶ more than 1% if convex (constant)
 - ▶ less than 1% if concave (increasing), $\xi_{q,e}|_{e=0} = 0$ and $\xi_{q,e}|_{e=1/\lambda} = 0.58$.
- ▶ The extra dollar the firm hides increase its probability of detection more than the previous dollar.
- ▶ For linear and exponential, the FOC bounds evasion *at* the expected value of T
 - ▶ For power, FOC bounds evasion *below* its expected value: At $k = 2$, $\frac{1}{\lambda\sqrt{3}} < \frac{2}{\lambda 3}$

Estimation and recommendation

- ▶ **Linear** and **exponential** carry a single parameter λ
- ▶ **Power** $(\lambda e)^k$ adds a parameter, curvature k

Do **linear** and **power**,

- ▶ Start with **linear** ($k = 1$). Linear is the $k = 1$ case of the power form (ceiling $(1+k)^{-1/k} = \frac{1}{2}$).
- ▶ Then, estimate power (λ, k) , imposing feasibility $\lambda e < (1+k)^{-1/k}$ for *every* firm. Test if $k = 1$.

The Counterfactual Question, Reframed

From “What’s Optimal?” to “What’s the Derivative?”

The natural first framing — “what purchases-tax credit rate Δ maximizes government revenue?” — presupposes an interior optimum exists. It doesn’t, in general, under this revenue formula (next slides).

What this section actually answers: holding sales-tax revenue and firms’ optimal materials/output decisions fixed at their observed values, and letting only evasion respond — what is the *derivative* of government tax revenue with respect to a change in the purchases-side tax credit rate, at the current state?

A well-defined, always-computable local quantity — not a claim that some Δ^* exists.

The Government's Problem (Two-Tax Model)

Only τ_P enters the firm's evasion FOC — τ_S never does. Shock $\tau_P \rightarrow (1 + \Delta)\tau_P \equiv \tilde{\tau}_{P,it}$:

$$R_i(\Delta; M_i) = \underbrace{\tau_{S,it} P_t Y_{it}}_{\text{sales-tax revenue, fixed}} - \tilde{\tau}_{P,it} [M_i + (1 - q(\tilde{e}_i(\Delta))) \tilde{e}_i(\Delta)]$$

M_i (true materials) held fixed for now — checked explicitly two slides from now.
Corner firms ($\tau_P = 0$): $R_i = \tau_{S,it} P_t Y_{it} - \tilde{\tau}_{P,it} M_{it}^*$, no detection term.

Closed Form: Evasion Under the Shock

Linear q , linear κ : new evasion is an exact linear function of the firm's own baseline evasion, with a firm-invariant intercept:

$$\tilde{e}_i(\Delta) = \frac{e_i}{1 + \Delta} + \frac{\Delta}{2\lambda(1 + \Delta)}, \quad \text{clipped at 0}$$

Mechanical consequence of κ linear in e : marginal cost is a firm-type constant, not a function of e — the FOC is linear, every firm's optimum shifts by the same absolute amount.

The Derivative, Holding M Fixed

Let $x \equiv 1 + \Delta$. For ANY q, κ (not just this linear spec) satisfying mild regularity — firm's own SOC holds, $\kappa' > 0$, detection not certain at the interior optimum — an envelope-theorem argument on the firm's own FOC gives:

$$\frac{dR}{dx} = -\tau_P \rho_t [M + L(e^*(x))] - \kappa'(e^*(x)) \frac{de^*}{dx} < 0 \quad \text{always,} \quad L(e) \equiv (1 - q(e))$$

A structural result, not a functional-form artifact: this revenue formula is a pure transfer between government and evading firms; nothing attaches a real resource cost to x . At the observed state ($\Delta = 0$), and everywhere else with M held fixed, raising the purchases credit always costs revenue on the margin — equivalently, cutting it always helps, on the margin, with materials/output held at their observed levels.

Robustness: Does the True Materials Level Actually Respond?

M was held fixed above — check it. The materials FOC (two-tax model) implies $M(x) \propto (1 - x\tau_P)^{-1/(1-\beta)}$, $M'(x) > 0$: a more generous credit lowers the net materials price, firms use more real materials. This makes $\tau_S P_t Y(x)$ respond too:

$$\frac{dR}{dx} = \rho_t M'(x) \cdot \frac{\tau_S - x\tau_P}{1 - \tau_S} - \tau_P \rho_t [M(x) + L(e^*(x))] - \kappa'(e^*(x)) \frac{de^*}{dx}$$

New term positive whenever $x\tau_P < \tau_S$ — in principle a real channel the fixed- M derivative misses.

Numeric Check: Real, but Negligible

$\Delta = 0$, median firm, sample medians ($\hat{\beta} = 0.387$, $\hat{\tau}_P = 0.088$, $\hat{\tau}_S = 0.089$), headline $\hat{\lambda} = 5.427 \times 10^{-7}$:

term	value
output-growth (new)	$\approx +1.8$
mechanical legit-credit	≈ -819
evasion-response	$\approx -80,655$

Evasion channel dominates $\$45,000 \times \$$ — driven by $1/(2\hat{\lambda}) \approx 921,000$, far larger than any M or e in the data. **Holding M fixed is not distorting the answer at current estimates** — the derivative's sign and rough magnitude are robust to this check. Full derivation: `Paper/sections/9999-tax-wedge.qmd`.

Scope

Both derivatives above describe the *local* response of revenue to Δ , around the observed equilibrium, holding $K, L, \omega, \varepsilon$ fixed throughout — a comparative-statics exercise, not a general-equilibrium optimal-tax result. What's estimated next (620-ELVIS.qmd, results in 650-stage2-prelim-results.qmd) is exactly this: the revenue effect of small changes in Δ around the current state, with evasion the only margin that moves.

Empirical Strategies

- ▶ Testing for the presence of tax evasion by industry
 - ▶ Robust to different specifications
- ▶ Regression
 - ▶ Useful to study tax evasion heterogeneity across firms and over time
- ▶ Deconvolution
 - ▶ Recover distribution of tax evasion and productivity
 - ▶ Estimation of unbiased Production Function

Data

Testing

Entropic Latent Variable Integration via Simulation (ELVIS)

An Alternative: ELVIS (Schennach (2014))

- ▶ The “Why not GMM” failure wasn’t fundamental — it was a *naive-averaging* bug: swapping in $\bar{\ell}_{it}(\lambda)$ silently drops the firm-specific reweighting
- ▶ Schennach (2014, *Econometrica*, “Entropic Latent Variable Integration via Simulation”) targets exactly this class: $E[g(U, Z, \theta)] = 0$ with U unobserved, g arbitrary — nonlinear, nonmonotone, nonseparable
- ▶ Her own canonical example (errors-in-variables: unobserved true regressor X^* , observed $X = X^* + \Delta X$) is structurally our M vs. M^*

Recasting the moment in M

Take M_{it} (true materials) — not ε_{it} — as the one unobservable. Everything else is a deterministic function of M and data:

$$\begin{aligned}e(M) &= M_{it}^* - M, & \varepsilon(M) &= \ln(M_{it}^*/M) - \mathcal{V}_{it}, \\ \omega(M) &= \tilde{\mathcal{W}}_{it} - (1 - \beta)\varepsilon(M)\end{aligned}$$

$$g(M, Z_{it}, \theta) \equiv \psi(M) = h(e(M)) - \delta_0 + \delta_1 \omega(M) - \delta_2 \omega(M)^2 \quad (12)$$

- ▶ Support $\mathcal{U}_{Z,\theta} = (M_{it}^* - \frac{1}{2\lambda}, M_{it}^*)$ — the feasibility ceiling $e < 1/(2\lambda)$ becomes the domain, not a complication (ELVIS explicitly allows θ -dependent support)
- ▶ Same ψ as before — recasting in M only changes *which variable ELVIS treats as unobserved*, not the moment itself

The entropic fix

Schennach's Theorem 2.1: the infinite-dimensional question “does *some* distribution of M given Z satisfy $E[g] = 0$ ” is **equivalent** to a finite-dimensional system in (θ, γ) :

$$\inf_{\mu} \|E_{\mu \times \pi}[g(M, Z, \theta)]\| = 0 \iff \inf_{\gamma \in \mathbb{R}^{d_g}} \|E_{\pi}[\tilde{g}(Z, \theta, \gamma)]\| = 0$$

$$\tilde{g}(Z, \theta, \gamma) \equiv \frac{\int g(M, Z, \theta) \exp(\gamma' g(M, Z, \theta)) d\rho(M|Z)}{\int \exp(\gamma' g(M, Z, \theta)) d\rho(M|Z)} \quad (13)$$

- ▶ The exponential tilt $\exp(\gamma' g(\cdot))$ **is** the missing reweighting — not assumed, but *solved for* as a Lagrange multiplier on the moment constraint
- ▶ $\rho(M|Z)$ is any user-chosen dominating measure — by her Remark 2.3, its choice has **zero effect** on the resulting estimate, only support matters

Estimation

Simulate $M^{(s)} \sim \rho(\cdot | Z_{it})$ (Metropolis) and form a ratio of averages:

$$\hat{g}(Z_{it}, \theta, \gamma) = \frac{\frac{1}{S} \sum_{s=1}^S g(M^{(s)}, Z_{it}, \theta) \exp(\gamma' g(M^{(s)}, Z_{it}, \theta))}{\frac{1}{S} \sum_{s=1}^S \exp(\gamma' g(M^{(s)}, Z_{it}, \theta))} \quad (14)$$

- ▶ Solve $(\hat{\theta}, \hat{\gamma})$ jointly by standard GMM on $E[\hat{g}(Z_{it}, \theta, \gamma)] = 0$
- ▶ λ identified the same way as before: sector–time variation in $\tau \rho_t$ shifting the evasion FOC's benefit
- ▶ 15 moments (finalized 2026-08-28), all built from independence assumptions the model already maintains — $\theta_{\text{smooth}} = (\delta_0, \delta_1, \delta_2, \sigma_\psi^2)$ (a new nuisance parameter, ψ 's own dispersion), λ gridded separately; full list in `Code/Rcpp/1200-stage2-elvis.cpp` / `CLAUDE.md`

ELVIS vs. MSL vs. plain MSM

	recovers f_ψ ?	needs f_ε ?	valid despite nonlinearity?
MSL (headline)	yes	yes	yes (by construction)
plain MSM	no	yes	no — naive-averaging bug
ELVIS	no	no — only support of M	yes (Thm 2.1)

- ▶ ELVIS's real payoff: stage 1's deconvolution of f_ε becomes **optional** for stage 2 — only $\hat{\beta}$ is needed to construct $\mathcal{V}, \tilde{\mathcal{W}}$
- ▶ $\hat{f}_\varepsilon$ can still be reused to build an efficient ρ (less simulation noise) — a computational choice, not an identification requirement

Going Further: One Unified Estimation

- ▶ If ELVIS works for stage 2 alone, it works for the *whole model* at once
- ▶ **There is only one true unobservable:** M_{it} — true materials, for every firm-period, corp and unincorporated alike
- ▶ No more $\mathcal{V}$, $\tilde{\mathcal{W}}$, no Jacobian, no density-transformation theorem — none of that machinery is needed once M itself is the latent
- ▶ Also sidesteps what made MSL “messier”: the AR(1) correction there needed the *conditional density* $f_{\omega_{it}|\omega_{it-1}}$ (i.e. f_{η} , recovered by Fourier deconvolution). ELVIS only needs **moment orthogonality** $E[\eta_{it}Z_{it}] = 0$ — no density of η at all

From M to ε

The materials FOC pins ε down exactly, given M :

$$\varepsilon_{it}(M_{it}) = \ln \beta - \ln \left(\frac{\rho_t M_{it}}{P_t Y_{it}} \right) \quad (15)$$

From M to ω — GNR's own trick, made exact

Substituting Equation 15 into the production identity

$\omega_{it} = y_{it} - \alpha_K k_{it} - \alpha_L l_{it} - \beta \ln M_{it} - \varepsilon_{it}$, the y_{it} terms **cancel exactly**:

$$\omega_{it}(M_{it}) = (1 - \beta) \ln M_{it} - \alpha_K k_{it} - \alpha_L l_{it} - \ln \beta + \ln \rho_t - \ln P_t \quad (16)$$

- ▶ This is **not information loss** — it's Gandhi-Navarro-Rivers's own $\Phi_{it} = y_{it} - \hat{\varepsilon}_{it}$ step, specialized to constant (Cobb-Douglas) elasticity: their Φ loses y_{it} the *same* way in this special case
- ▶ Standard proxy-variable logic (Olley-Pakes/Levinsohn-Petrin/ACF/GNR): under the timing assumption, optimal m is a monotonic function of ω *alone* — so m already perfectly proxies ω . y_{it} 's only job is revealing ε_{it} (via the share equation); once that's done, y_{it} has nothing further to say about ω
- ▶ y_{it} is used exactly **once** — inside $\varepsilon_{it}(M_{it})$

The AR(1) moment

Raw form (before substituting the FOC) — note the correct coefficients:

$$\begin{aligned} & y_{it} - \beta \ln M_{it} - \alpha_K k_{it} - \alpha_L l_{it} - \gamma_0 \\ & \quad - \gamma_1 (y_{it-1} - \beta \ln M_{it-1} - \alpha_K k_{it-1} - \alpha_L l_{it-1}) \\ & = \eta_{it} + \varepsilon_{it} - \gamma_1 \varepsilon_{it-1} \end{aligned}$$

Since ε is itself just a function of M (Equation 15), substitute Equation 16 directly — no ε left anywhere:

$$\eta_{it}(M_{it}, M_{it-1}) = \omega_{it}(M_{it}) - \gamma_0 - \gamma_1 \omega_{it-1}(M_{it-1}) \quad (17)$$

Evasion moments, by firm type

Unincorporated (interior FOC), same as before:

$$e_{it}(M_{it}) = M_{it}^* - M_{it},$$
$$\psi_{it}(M_{it}) = h(e_{it}(M_{it})) - \delta_0 + \delta_1 \omega_{it}(M_{it}) - \delta_2 \omega_{it}(M_{it})^2$$

Corp — relaxed from a hard constraint to a centering moment:

$$E[e_{it}(M_{it}) \mid \text{Corp}_i] = 0 \quad \text{or} \quad \text{Med}[e_{it}(M_{it}) \mid \text{Corp}_i] = 0$$

- ▶ No longer “corp never evade” imposed *outside* the estimation — corp can evade a little, just centered at zero, matching what the data already show (near-zero, imprecise evasion at both tails)

Unincorporated — evade on average, as an inequality:

$$E[e_{it}(M_{it}) \cdot \mathbb{1}(\text{Unincorp}_i) - U_2] = 0, \quad U_2 \geq 0$$

- ▶ ELVIS’s own device for inequalities (Example 1.3 in Schennach (2014)) — no separate technique needed

The full moment system

Moment	Object	Identifies	Instruments
Materials FOC	$\varepsilon_{it}(M_{it})$	β (mainly via corp)	constant / sector
AR(1) innovation	$\eta_{it}(M_{it}, M_{it-1})$	$\alpha_K, \alpha_L, \gamma_0, \gamma_1$	k_{it}, l_{it} , lagged m^*
Evasion FOC (unincorp)	$\psi_{it}(M_{it})$	$\lambda, \delta_0, \delta_1, \delta_2$	sector-time $\times \tau \rho_t$
Corp evasion	$e_{it}(M_{it})$	pins $M_{it} \approx M_{it}^*$ for corp	— (centering)
Unincorp evasion sign	$e_{it}(M_{it}) \cdot \mathbb{1}(\text{Unincorp})$	tightens the identified set	— (inequality)

- ▶ All firm-periods, corp and unincorporated, enter **jointly** — no data thrown away at any stage
- ▶ One draw of M_{it} (and its lag) feeds the *whole* moment vector — not double-counting, this is standard for any GMM system with multiple moments per observation
- ▶ **Before committing:** verify the stacked Jacobian $\partial E[g]/\partial \theta_{\text{full}}$ has full column rank numerically — algebra confirms the construction is sound, not that every parameter is well-identified in finite samples

More moments from assumptions we already make

Every independence assumption already maintained — but never yet written as a moment — adds an (almost free) row to g :

Assumption	New moment
$\varepsilon \perp \psi$	$E[\varepsilon_{it}(M_{it})\psi_{it}(M_{it})] = 0$
$\omega \perp \varepsilon$	$E[\omega_{it}(M_{it})\varepsilon_{it}(M_{it})] = 0$
$\psi \perp \omega$	$E[\psi_{it}(M_{it})\omega_{it}(M_{it})] = 0$ — overidentifying check on the U-shape
$\varepsilon_{it} \perp \varepsilon_{it-1}$	$E[\varepsilon_{it}(M_{it})\varepsilon_{it-1}(M_{it-1})] = 0$ — free, M_{it-1} already drawn
$\varepsilon_{it-1} \perp \psi_{it}$	$E[\varepsilon_{it-1}(M_{it-1})\psi_{it}(M_{it})] = 0$ — free, same draw

- ▶ Formal license: Schennach (2014)'s own supplement, Prop. B.1 (conditional-mean-zero $\square$ countable unconditional moments $E[Ye^{i\xi X}] = 0$) and Prop. B.2 (independence, not just uncorrelatedness, factors the same way through characteristic functions)
- ▶ Her own examples do this in practice — Example 1.5 (closest analog here) adds a cross-moment between its two error terms; the interval-regression example adds a variance/homoskedasticity moment purely to narrow the bounds
- ▶ **Not free:** the confidence-region critical value is $\chi^2_{d_g, \alpha}$, d_g =number of moments — every row raises the bar. Recommended: add these six first, treat the full characteristic-function grid as a later robustness pass

What about the counterfactual?

MSL's revenue integral weights by $f_{\psi}(\psi(\varepsilon))J_i(\varepsilon)f_{\omega}(\omega(\varepsilon))f_{\varepsilon}(\varepsilon)$ — needs the full recovered density, which ELVIS doesn't deliver.

- ▶ ELVIS's own machinery for exactly this situation: Schennach's device for bounding a functional of the unobservable that isn't already one of the estimating moments — her worked example is literally recovering $E[U]$ this way (Schennach (2014), p.1250)
- ▶ Reusing the tilted density from the *main* estimation as if it were “the” posterior would give a number, but understate uncertainty — that tilted distribution is only the *entropy-maximizing* member of the set of distributions consistent with the moments, not necessarily the true one (she is explicit about this). The honest version re-solves for the functional's own identified set

Implementing it: an auxiliary moment

Hold $\hat{\theta}$ fixed (already estimated — the counterfactual is a forward simulation, not new estimation).
Reuse the closed-form counterfactual evasion FOC already established:

$$\tilde{e}_i(\tilde{\tau}) = \tilde{e}(\tilde{\tau}; \omega_i(M_i), \psi_i(M_i)) = \frac{\tilde{\tau}\rho_t - C(\omega_i, \psi_i)}{2\lambda\tilde{\tau}\rho_t}, \quad C(\omega, \psi) \equiv \mathbf{e}^{\delta_0 - \delta_1\omega + \delta_2\omega^2 + \psi}$$

(true M_i unchanged under the counterfactual — materials FOC doesn't depend on τ , per the model's own maintained assumptions) and form firm i 's counterfactual revenue as a function of the *same* latent M_i :

$$R_i(\tilde{\tau}; M_i) = \tilde{\tau} \left[P_t Y_{it} - \rho_t M_i - (1 - q(\tilde{e}_i(\tilde{\tau}))) \rho_t \tilde{e}_i(\tilde{\tau}) \right]$$

For a candidate $\tilde{\tau}$, introduce an auxiliary parameter $\tilde{\theta}(\tilde{\tau}) \equiv \frac{1}{N} \sum_i E[R_i(\tilde{\tau}) \mid Z_i]$ and **append its own moment** to the system:

$$E[R_i(\tilde{\tau}; M_i) - \tilde{\theta}] = 0$$

with θ fixed at $\hat{\theta}$ — only $\tilde{\theta}$ and an *extended* γ (one more component, for the new moment) are free.
Same simulated-draw, ratio-of-averages machinery as before (Equation 14), just evaluated on the extended moment vector.

Two Practical Corrections

θ is held fixed, not reprofiled, at every candidate cell — only γ is free. Letting θ move too lets a weakly-identified λ silently rationalize almost any candidate R (confirmed empirically: $\hat{\lambda}$ swung $56\times$ across cells when left free) — answering “does *some* enforcement regime make this R possible,” not the intended question. AK2020’s own counterfactual (Appendix_F/) does the same: their structural parameter is a hardcoded constant throughout, only γ is passed to the optimizer.

The revenue moment itself is control-variate adjusted, not raw: R_i regressed against the firm’s own already-observed $t1_i/pgdp_i$ ($R^2 = 0.999$) removes size-driven between-firm heterogeneity from the moment’s sampling variance (SE cut $\$40\times\$$) while leaving its mean unchanged by construction — what actually turned a vacuous test into a real, closed confidence set.

Point or set — decided by the procedure, not assumed

For a fixed $\tilde{\tau}$, grid over **candidate revenue values** $\tilde{\theta}$ (not over θ this time):

1. For each candidate $\tilde{\theta}$, solve the inner problem — maximize the objective (EL, or the CUE-type quadratic $-\frac{1}{2}\hat{g}'\hat{V}^{-1}\hat{g}$) over the extended γ , with the $R_i(\tilde{\tau}; M_i) - \tilde{\theta}$ moment included
 2. That maximum is a criterion function $\hat{L}_n(\tilde{\theta}; \tilde{\tau})$ — exactly the same construction used for θ itself
 3. **Profile against the unrestricted model:** $\hat{Q}_n(\tilde{\theta}; \tilde{\tau}) = -(\hat{L}_n(\tilde{\theta}; \tilde{\tau}) - \sup_{\theta', \gamma} \hat{L}_n(\theta', \gamma))$ — subtracting the global max, not using $\hat{L}_n$ raw, is what gives the efficiency gain this construction is built for
 4. Region: $\{\tilde{\theta} : n\hat{Q}_n(\tilde{\theta}; \tilde{\tau}) \leq c_\alpha\}$, $c_\alpha = \chi_{d_g, \alpha}^2$ — Theorem F.1's closed-form conservative critical value (Schennach 2014, online supplement, Appendix F). This is the test we report; a min-subtracted CHT-style alternative was useful for guiding earlier searches but isn't the textbook construction, so it isn't reported here.
- ▶ Repeat across a grid of $\tilde{\tau}$: **the shape decides**. A sharp trough at every $\tilde{\tau} \rightarrow$ the Laffer curve comes out as a single curve, point-identified. A flat plateau at some $\tilde{\tau} \rightarrow$ a genuine band there — the model just doesn't pin expected revenue down more tightly at that rate
 - ▶ We do **not** commit to point vs. set beforehand — both come out of the identical grid-and-criterion-function procedure; the width of the trough is the answer, not a modeling choice
 - ▶ $\tilde{\tau}^*$ (or a *range* of $\tilde{\tau}^*$, if the argmax band is itself wide) follows directly by reading off where the revenue band peaks
 - ▶ Convergence rate itself depends on the regime: root- n if the objective is exactly flat over the identified set (“degeneracy,” her interval/censored examples), only $\sqrt{\ln n/n}$ otherwise — a property of the fitted objective to check, not to assume

Stage-2 ELVIS: Preliminary Results

Implementation

I use 9 moments: $\theta_{\text{smooth}} = (\lambda, \delta_0, \delta_1, \delta_2)$, 9 moments ($d_g = 9$), nuisance parameters (γ, δ_0) :

$$g(M, Z, \theta_{\text{smooth}}) = \begin{pmatrix} \psi(M, Z) \\ \varepsilon(M, Z) \\ \psi(M, Z) \ln M \\ \psi(M, Z) \omega(M, Z) \\ \psi(M, Z) \omega(M, Z)^2 \\ \varepsilon(M, Z) \ln M \\ \varepsilon(M, Z) e(M, Z) \\ \varepsilon(M, Z) \omega(M, Z) \\ h'(e, \lambda) \varepsilon(M, Z) \end{pmatrix}$$

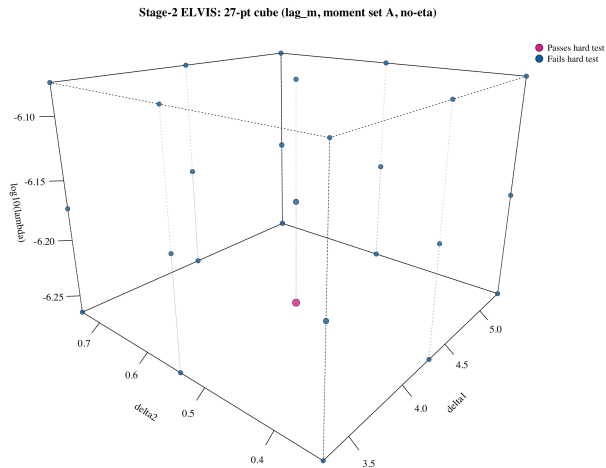
Strategy

- ▶ First, grid search over λ , profiling over $(\delta_0, \delta_1, \delta_2)$, to find the optimal λ and a rough (δ_1, δ_2) region
- ▶ Then, a 2D grid search over (δ_1, δ_2) around the best $(\lambda, \delta_1, \delta_2)$ from the λ search, profiling over (λ, δ_0)
- ▶ Finally, a joint 3D search over $(\lambda, \delta_1, \delta_2)$, starting at the best point from the δ 's grid with only δ_0 free, to refine the best point
- ▶ γ 's are always free in all searches

Implementation Details

- ▶ **Objective:** continuous updating estimator (CUE) quadratic,
 $\hat{L}_n(\theta, \gamma) = \frac{1}{2} \hat{g}' \hat{\Omega}^- \hat{g}$, $\hat{\Omega}^-$ an eigenvalue-truncated generalized inverse (following Aguiar and Kashaev 2020)
- ▶ **Sampler:** per-firm Metropolis chain, `n_burn=1000`, `n_keep=3000`, deterministic per-firm RNG reseed
- ▶ **Solver:** Nelder-Mead (BOBYQA got stucked in local minima), unbounded γ (a large/degenerate γ might be an expected solution, (Schennach 2014)), two-pass refinement
- ▶ **Seeding:** every grid point seeded independently from the same validated starting point

$(\lambda, \delta_1, \delta_2)$ -Search, m_{it-1}^*



Ball size ~ p-value of the HARD test (bigger=more significant). Pink=passes, blue=fails. Min at d1=4.3 d2=0.54 lambda=5.43e-07.

Headline Estimates

Table 1

	m_{t-1}^* ^a	$\tilde{W}_{t-2}$ ^b
$\hat{\lambda}$	5.43e-07	5.43e-07
$\hat{\delta}_0$	3.464	10.206
$\hat{\delta}_1$	4.300	7.460
$\hat{\delta}_2$	0.540	0.972
$\hat{\omega}^* = \hat{\delta}_1/2\hat{\delta}_2$	3.981	3.839
$\max \gamma $	2.644	1.863
TS_{hard}	13.860	25.867
$\chi^2_{9,.95}$	16.919	16.919
Result	Passes	Fails

^a m_{it-1}^* 's point estimate refined from a joint 3D $(\lambda, \delta_1, \delta_2)$ grid (27 cells).

^b $\tilde{W}_{it-2}$'s point estimate from λ -only grid search.

Interpreting the Estimates

- ▶ Probabilities of detection are low, average audit-triggering threshold $1/(2\hat{\lambda}) \approx \$920,000$, while median reported materials $\text{Med}[M^*] \approx \$8,414$, the 95th percentile firms \$128,154, the 99th \$360,676, and the largest (trimmed) firm \$701,130
- ▶ The cost function displays the expected U-shape, both $(\hat{\delta}_1, \hat{\delta}_2)$ display correct signs.
 - ▶ $\hat{\delta}_1 < 0$ and $\hat{\delta}_2 > 0$ the evasion cost function is decreasing in productivity up to $\hat{\omega}^*$, then increasing again
 - ▶ $\hat{\omega}^* = 3.98 (m_{it-1}^*)$ and $3.84 (\tilde{\mathcal{W}}_{it-2})$, the 86th (m_{it-1}^*) and 75th percentile $(\tilde{\mathcal{W}}_{it-2})$ (forward simulation)

How the Test Works

Test statistic at a candidate θ : $TS(\theta) = 2n\hat{L}_n(\theta)$, profiled over γ (and any nuisance parameters).

- ▶ Reported test: reject θ if $TS(\theta) > \chi^2_{d_g, .95}$ (d_g = number of moments)
(**Theorem F.1**, Appendix F. Schennach 2014)

What Detection Probability Does This Imply?

- ▶ Mean $\gg$ Median for both instruments; a right-skewed evasion distribution
- ▶ P99 $\rightarrow$ Max: detection probability jumps from 1-2% at the 99th percentile, to ~37% for the most aggressive evader in the (0.5%-trimmed) sample.
- ▶ A handful of extreme evaders pull $E[e]$ **up** and the fitted $\hat{\lambda}$ **down**

Table 2

Stat	m_{t-1}^*		$\tilde{W}_{t-2}$	
	e	$q(e)$	e	$q(e)$
Mean	1,946	0.11%	3,171	0.17%
Median	495	0.03%	687	0.04%
P95	6,912	0.38%	12,170	0.66%
P99	20,099	1.09%	37,488	2.03%
Max	694,645	37.70%	683,975	37.12%

Implied detection probability $q(e) = \hat{\lambda} \cdot e$ across the evasion distribution, using forward-simulation estimates.

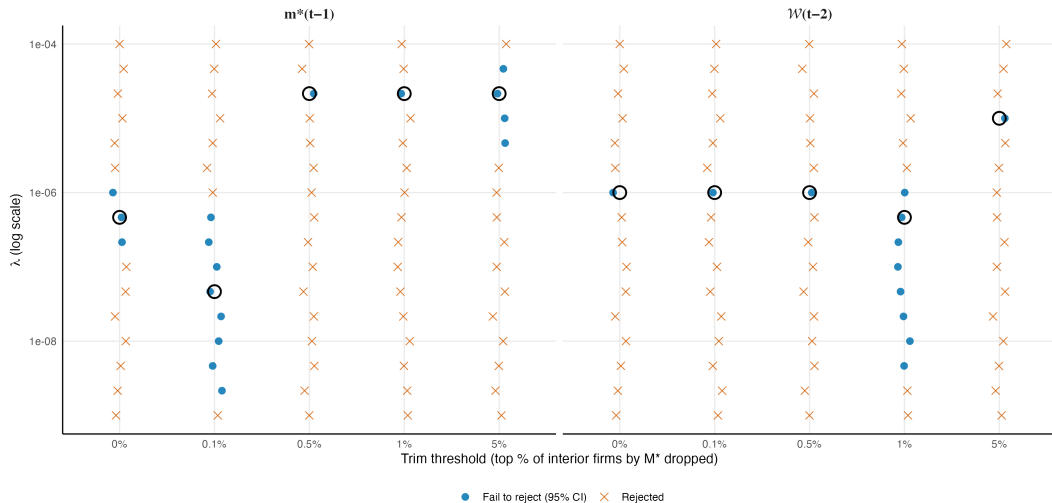
Trimming Sample

Since few aggressive evaders were pushing the estimates of λ down, I trimmed the sample at different percentiles of M^* to test sensitivity of the estimates.

- ▶ At 0.5% trim, λ seems to stabilize before jumping or start waving too much
- ▶ δ 's looked stable across the trim range, confirming that they were fitting the curvature of evasion, and lambda the level accross firms

Stage-2 lambda grid: test-inversion pass/fail at every tested point

Re-swept with the fixed (floored + exp-bounded) h' score moment. Open circle = argmin within that trim level.
 Reject if $TS=2n[Lhat-min(Lhat)] > \chi^2_{\{8,0.95\}}=15.51$. x-axis is discrete/equally-spaced, not proportional to trim %.



Notes for Discussion

- ▶
- ▶ m_{it-1}^* **is the instrument to lead with:** stable $\hat{\delta}_1, \hat{\delta}_2$ across every search run so far, and the only instrument/grid combination to ever clear the conservative hard test
- ▶ $\tilde{\mathcal{W}}_{it-2}$ **confirms the same qualitative story but is the weaker instrument:** same U-shaped sign on $\hat{\delta}_1, \hat{\delta}_2$, similar $\hat{\omega}^*$, but doesn't clear the hard test at its own best point, and hasn't been run through the joint grid yet
- ▶ **A first joint 3D grid is in (27 cells, coarse spacing)** — not yet a full confidence region (only 1 tested cell clears the hard test), but real evidence for the fixed $\hat{\theta}$ used in the counterfactual

λ Search

- ▶ λ bounds derived directly from the data: the smallest λ for which not even the largest firm's evasion is ever constrained, up to the largest λ for which even the smallest firm is constrained
 - ▶ FOC: $\lambda < \frac{1}{2e}$ approximated with $M^* = M + e$, assuming all is evasion, then the relevant range for λ is $[1/(2 \max(M^*)), 1/(2 \min(M^*))]$
 - ▶ During estimation, λ never hits the bounds
- ▶ 16-point log-spaced grid (coarse pass, then a finer pass bracketing the region of interest)
- ▶ **Sample:** 32,232 unincorporated firm-periods (3,340 at the $\tau_P = 0$ corner), top 0.5% trimmed by M^* ; two productivity-instrument choices for $(\alpha_K, \alpha_L, \beta)$

Laffer Curve (Counterfactual Question)

Increasing Efficiency in the Counterfactual Exercise

Motivation and theoretical derivation: 630-counterfactual-question.qmd.

Implementation: 620-ELVIS.qmd.

Goal: use a control variate so the counterfactual test of candidate revenue levels is more *efficient* — same target, less noise.

Why: testing candidate revenue R against the data was almost powerless. Revenue is dominated by *firm size* — large firms pay vastly more sales tax than small ones — and that size variation has nothing to do with evasion. It's noise swamping the signal we actually care about.

How: each firm's own sales tax paid, $t1_i$, is fully observed and predicts revenue almost perfectly ($R^2 \approx 0.999$). Subtract off the size-driven, predictable part of revenue before testing — the classic control-variate variance-reduction technique (Hendry 1984), the same logic behind ANCOVA / CUPED adjustment in modern experiments.

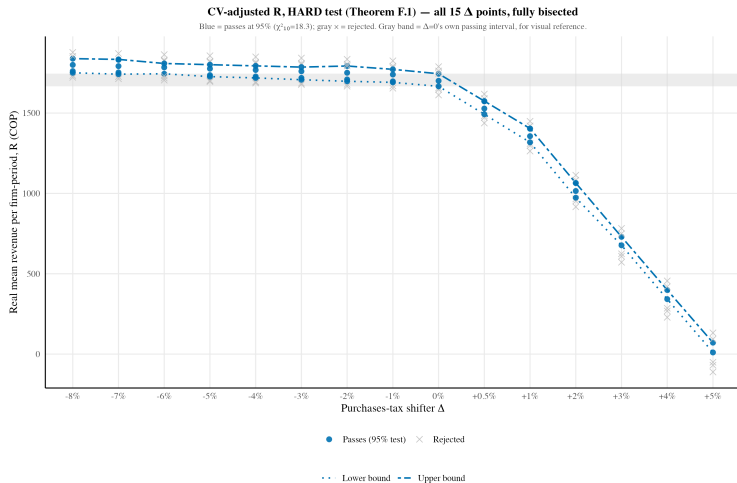
Interpretation: the estimator stays **consistent and unbiased for the same expected revenue** $E[R_i(\Delta)]$ — nothing about what a candidate R means, or how to read a confidence interval, changes. The correction only strips out noise the estimator didn't need; the same population quantity is estimated **more efficiently** — a tighter interval around the same target, not a different one.

Control-Variate Moment — Details

$$g_{R,i}^{\text{adj}} = \left[\frac{R_i(\Delta)}{pgdp_i} - \hat{\beta} \frac{t1_i}{pgdp_i} \right] - [R - \hat{\beta} \mu_c]$$

- ▶ $\hat{\beta} = 0.9988$ (OLS, pooled across tested Δ 's), $\mu_c = E[t1_i/pgdp_i]$ — both fixed, precomputed constants, never new (θ, γ) decision variables
- ▶ $E[g_{R,i}^{\text{adj}}] = E[g_{R,i}]$ exactly for *any* $\hat{\beta}$ — verified to 6+ sig figs numerically, not just algebraically
- ▶ Beats industry $\times$ year TWFE decisively (38-41x vs. 2.5x), zero added dimensions, no small-cell risk (100/318 TWFE cells have < 30 obs)

Testing the Laffer Curve



Crossing found: Delta=-8% is the smallest tax cut statistically distinguishable from Delta=0 (sequential search, see Research-log(log.md)). Cuts -1% to -7% still overlap the Delta=0 band. Increases $\geq +0.5\%$ separate cleanly.

Every point is a real candidate revenue level, tested against the data (95% confidence, gray band = current policy's own interval)

Headline Result

- ▶ A tax **increase** is bad news almost immediately: even a **0.5% increase** is already statistically distinguishable from current policy
- ▶ A tax **cut** helps too, but the gain only becomes statistically distinguishable once the cut reaches about **8%** — much larger than intuition suggests
- ▶ Both results come from a formal statistical test, not a point estimate

Table 1: Statistically distinguishable revenue impact: an 8% cut vs. a 0.5% increase (hard test, 95%)

Scenario	Tax change	Revenue CI (real COP)	Statistically guaranteed change	Point-estimate change
Baseline (current policy)	0.0%	[1666, 1745]	–	–
Tax cut (smallest statistically distinguishable)	-8.0%	[1751, 1839]	+0.35% to +10.34%	+5.84%
Tax increase (smallest tested)	+0.5%	[1492, 1575]	-14.45% to -5.50%	-10.15%

Why

Linear q + linear $\kappa \Rightarrow$ new evasion is exactly linear in the firm's own baseline evasion, with an intercept identical across every firm:

$$\tilde{e}(\Delta) = \frac{e_i + \Delta/(2\lambda)}{1 + \Delta}$$

Whether this looks like *everyone moves together* or *heterogeneous, firm-by-firm* is a horse race between e_i and $\Delta/(2\lambda)$. Enforcement is estimated to be weak ($\hat{\lambda}$ tiny), so $\Delta/(2\lambda)$ wins: a tax increase shifts every firm by roughly the same large amount, while a cut removes evasion firm-by-firm — small evaders reach zero first, so the aggregate response is gradual instead of a jump.

Estimating Tax Evasion by Deconvolution

Objective

- ▶ Empirical exercise to motivate structural model

What can be deconvolved and what can't

Convolution (Meister, 2009)

The density of the sum of two **independent** random variables is equal to the **convolution** of the densities of both addends; hence

$$f_V = \int f_\varepsilon(V - u)f_u(u)du$$

Under $\varepsilon \perp\!\!\!\perp M$, e and $e \perp\!\!\!\perp M$, with known f_ε and $u = \ln\left(\frac{M+e}{M}\right)$. Also $M^* = M + e$, and $V = u + \varepsilon$.

Then, I can deconvolve:

► f_u : $f_V = \int f_\varepsilon(V - u)f_u(u)du$

I don't need f_e , actually

What deconvolution recovers is the **overreporting log-ratio**,

$$u_{it} \equiv \ln \left(\frac{M_{it}^*}{M_{it}} \right) = \ln \left(1 + \frac{e_{it}}{M_{it}} \right),$$

- ▶ the *share of overreported over true inputs*
- ▶ the same object discussed previously, now given an explicit structural interpretation
- ▶ Since it's a ratio, it's adjusted by the firm's true materials level and hence informative about the intensity of tax evasion
- ▶ Had I recovered e , I would have to normalize somehow to compare firms of different levels of material utilization

Getting rid of the log

If I'm interested in only the density of the ratio, I can use a **density transformation**

Theorem 1.7.1 (Hogg et al. 2019): Let X be a continuous random variable with pdf $f_X(x)$ and support $\mathcal{S}_X$. Let $Y = g(X)$ where $g(x)$ is a one-to-one *[monotonic] differentiable* function on the support of X , $\mathcal{S}_X$. Denote the inverse of g by $x = g^{-1}(y)$ and $dx/dy = d[g^{-1}(y)]/dy$. Then, the pdf of Y is given by

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{dx}{dy} \right|, \text{ for } y \in \mathcal{S}_Y$$

where the support of Y is the set $\mathcal{S}_Y = \{y = g(x) : x \in \mathcal{S}_X\}$

Getting density of the ratio

Therefore, with $X = u$ and $g^{-1}(y) = \ln(1 + y)$

$$f_{\frac{e}{M}}(y) = f_u(\ln(1 + y)) \left| \frac{1}{1 + y} \right| \quad (18)$$

Informal proof

If $U = g(R)$ for a strictly monotonic differentiable g , then

$$f_R(r) = f_U(g(r))|g'(r)|$$

Proof (via CDFs). Take g strictly increasing (your case). Then

$$F_R(r) = P(R \leq r) = P(g(R) \leq g(r)) = P(U \leq g(r)) = F_U(g(r)),$$

using that g increasing preserves the inequality direction.

Differentiate both sides in r :

$$f_R(r) = \frac{d}{dr}F_U(g(r)) = f_U(g(r))g'(r).$$

If g were decreasing instead, the inequality flips ($P(U \geq g(r)) = 1 - F_U(g(r))$), and differentiating introduces a minus sign that exactly cancels the sign of $g'(r)$, hence $|g'(r)|$ of the general formula.

Everything goes through unchanged conditional on $Z = z$ (just work with $F_{R|Z}(\cdot | z)$, $F_{u|Z}(\cdot | z)$ throughout).

Discussion

Deconvolution estimates the unconditional f_u because of how we set it up

- ▶ From f_u , we can learn the probability of a firm of overreporting u ,
 $P(U < u) = \int_0^u f_u(t)dt$, or
 - ▶ the expected overreporting $\mu_u = \int u f_u(u)du$
- ▶ This is the unconditional probability, the same for all firms regardless of their characteristics
- ▶ We might be interested in the *probability of overreporting conditional* on the firm size

What if we are interested in the conditional density?

$$\begin{aligned} f_{V|Z}(V | Z) &= \frac{f_{V,Z}(V, Z)}{f_Z(Z)} \\ &= \int f_{V|u,Z}(V | u, Z) f_{u|Z}(u | Z) du \\ &= \int f_{\varepsilon}(V - u) f_{u|Z}(u | Z) du \end{aligned}$$

using

- ▶ $f_{V|u,Z}(V | u, Z) = f_{\varepsilon|u,Z}(V - u | u, Z)$ — $V = u + \varepsilon$ conditional on u
- ▶ $\varepsilon \perp (u, Z)$ **jointly**, so $f_{\varepsilon|u,Z} = f_{\varepsilon}$

with $u \not\perp Z$ (otherwise conditioning on Z would be pointless).

Estimating the Conditional distribution

- ▶ **Discrete/bucketed** Z (e.g. quintiles of K_{it} or L_{it}): the same 1-D deconvolution separately within each bucket.
 - ▶ We've already looked into $E[u|Z]$
- ▶ **Continuous** Z : nonparametric deconvolution with a continuous covariate
 - ▶ Literature: $f_{u|V}(u|V)$ (Wang and Ye 2015), $E[Z|u]$ (Fan and Truong 1993), and $f_{Z|u}$ (Huang and Zhou 2020)
 - ▶ $f_{u|Z}$: feasible using characteristic functions and kernels
- ▶ Ideally, what I want is $f_{e|\omega}$, but that's what f_{ψ} is, the object I will obtain with the structural approach

Conditional Deconvolution — Backup

What About the Full Distribution?

- ▶ So far: $E[u_{ijt} \mid Z_{ijt}]$, and only at 10 decile bins of Z
- ▶ Natural next question: the whole conditional density $f_{u|Z}(u \mid z)$, for Z continuous
 - ▶ Is the size gradient in the *level* of evasion, its *spread*, or its *shape*?

Aside: How Deconvolution Works

We never observe u_{ijt} directly — only $V_{ijt} = u_{ijt} + \varepsilon_{ijt}$, a “blurred” version of it

- ▶ A histogram of V mixes the true shape of u with the noise ε — the density of a sum of independent things is the *convolution* of their densities, $f_V = f_u * f_\varepsilon$
- ▶ Convolutions are hard to undo directly, but they become simple *multiplication* after a Fourier transform (the characteristic function):

$$\phi_V(t) = \phi_u(t) \phi_\varepsilon(t) \quad \implies \quad \phi_u(t) = \frac{\phi_V(t)}{\phi_\varepsilon(t)}$$

- ▶ Estimate ϕ_V from the data, divide by the already-known ϕ_ε , transform back $\rightarrow \hat{f}_u$
- ▶ This is exactly the (unconditional) trick behind the first-stage results (Carroll and Hall 1988; Stefanski and Carroll 1990)

Conditional Version: Add a Local Window

Same trick, done separately for each value of Z :

- ▶ Instead of using *all* firms to build one blurred distribution, use only firms with Z_i close to z — a moving local window, weighted by a kernel K
- ▶ Un-blur *that* local distribution the same way as before

$$\hat{\phi}_{V|Z=z}(t) = \frac{\sum_i K\left(\frac{Z_i - z}{h}\right) e^{itV_i}}{\sum_i K\left(\frac{Z_i - z}{h}\right)}$$

$$\Rightarrow \hat{f}_{u|Z=z}(u) = \frac{1}{2\pi} \int e^{-itu} \frac{\hat{\phi}_{V|Z=z}(t)}{\phi_\varepsilon(t)} \omega(t) dt$$

- ▶ $\omega(t)$: a damping weight, standard in any deconvolution, to keep the division numerically stable
- ▶ Valid because $\varepsilon \perp (u, Z)$: within any Z -window, the noise is still independent of u — the same assumption used throughout this section

Is This Standard?

- ▶ (Fan and Truong 1993), already cited on the “is it feasible?” slide, addresses the *mirror-image* problem: a noisy covariate and a clean outcome (see also (Huang and Zhou 2020) for the conditional-density version of that same direction)
- ▶ Ours runs the other way — clean Z , noisy u — which is why the conditional *mean* never needed correcting: $\varepsilon \perp Z \Rightarrow E[V \mid Z] = E[u \mid Z]$
- ▶ Only the *density* needs deconvolving, and the tool is just the classical deconvolution kernel (Carroll and Hall 1988; Stefanski and Carroll 1990), applied inside a local window — a natural combination, not (to our knowledge) a named result in the literature

Do We Need the Full Density?

- ▶ The conditional mean already carries the headline story: evasion rises then falls with size — going further only pays off if it answers a question the mean can't
- ▶ The one moment that would matter: the conditional **variance** $\text{Var}(u \mid Z = z)$
 - ▶ Does “large firms don't evade” mean (a) *every* large firm individually evades a little — tight distribution near zero, like the corporations — or (b) a mix of some firms still evading heavily and others near zero that cancel out in the mean?
 - ▶ (a) vs. (b) is a real audit-targeting question: under (a), size alone is a good targeting rule at the top; under (b) it isn't — some large firms are still hiding substantial evasion behind the average
- ▶ If pushed further: the full nonparametric conditional deconvolution (previous slides) is the general method; cumulant/method-of-moments matching is a cheap over-identification check on the variance estimate, not a substitute for it

Appendix

Appendix

Naive MSM and MSL

Recall: Two Identities From the Model

$$\mathcal{V}_{it} \equiv \ln \left(\frac{\rho_t M_{it}^*}{P_t Y_{it}} \right) - \ln(\beta) = \ln \left(\frac{M_{it}^*}{M_{it}} \right) - \varepsilon_{it} \quad (19)$$

$$\tilde{\mathcal{W}}_{it} \equiv \mathcal{W}_{it} - \alpha_K k_{it} - \alpha_L l_{it} = \omega_{it} + (1 - \beta)\varepsilon_{it} \quad (20)$$

Solving for M and e

NEW From Equation 19, I can *algebraically solve* for true inputs as function of observed variables and an the error term,

► $M_{it} = M_{it}^* \mathbf{e}^{-\mathcal{V}_{it} - \varepsilon_{it}}.$

Therefore,

$$\begin{aligned} e_{it} &= M_{it}^* - M_{it} \\ &= M_{it}^* \left(1 - \mathbf{e}^{-\mathcal{V}_{it} - \varepsilon_{it}} \right) \end{aligned}$$

- ε_{it} is not observed firm-by-firm, only its distribution f_ε
- not a literal point recovery of e_{it} , but the map that could be evaluated for some draw of $\varepsilon^{(s)} \sim \hat{f}_\varepsilon$

Recast in terms of one unobservable

I can do the same and solve for ω_{it} from Equation 20

For each value of ε everything else is pinned down:

$$e_{it}(\varepsilon) = M_{it}^* (1 - \mathbf{e}^{-\mathcal{V}_{it} - \varepsilon}) \quad (21)$$

$$\omega_{it}(\varepsilon) = \tilde{\mathcal{W}}_{it} - (1 - \beta)\varepsilon \quad (22)$$

$$\psi_{it}(\varepsilon) = h(e_{it}(\varepsilon)) - \delta_0 + \delta_1 \omega_{it}(\varepsilon) - \delta_2 \omega_{it}(\varepsilon)^2 \quad (23)$$

Drawing from f_ε

Substituting Equation 21 and Equation 22 into Equation 23, and simplifying

$$\begin{aligned}\psi_{it} + \delta_1(1 - \beta)\varepsilon = & \ln(\tau\rho_t) + \ln(1 - 2\lambda M_{it}^*(1 - \mathbf{e}^{-\mathcal{V}_{it} - \varepsilon})) \\ & - \delta_0 + 2\delta_2(1 - \beta)\tilde{\mathcal{W}}_{it}\varepsilon - \delta_2(1 - \beta)^2\varepsilon^2 \\ & + \delta_1\tilde{\mathcal{W}}_{it} - \delta_2\tilde{\mathcal{W}}_{it}^2\end{aligned}\tag{24}$$

Taking conditional expectation

Assuming $\psi \perp Z$, $\varepsilon \perp Z$, and $\omega \perp \varepsilon$.

$$\varepsilon \perp Z \Rightarrow E[\delta_1(1 - \beta)\varepsilon \mid Z] = \delta_1(1 - \beta)E[\varepsilon \mid Z] = 0$$

$$\begin{aligned} 0 = & \ln(\tau\rho_t) + E[\ln(1 - 2\lambda M_{it}^*(1 - \mathbf{e}^{-\mathcal{V}_{it} - \varepsilon})) \mid Z] \\ & \underbrace{- \delta_0 - \delta_2(1 - \beta)^2\sigma_\varepsilon^2}_{\text{Constant (known from stage 1)}} \\ & + \delta_1\tilde{\mathcal{W}}_{it} - \delta_2\tilde{\mathcal{W}}_{it}^2 \end{aligned} \tag{25}$$

Why not simulated GMM with instruments?

Everything in Equation 25 is data and parameters, **except**:

$$\begin{aligned}\ell_{it}(\varepsilon; \lambda) &\equiv \ln(1 - 2\lambda M_{it}^* (1 - \mathbf{e}^{-\mathcal{V}_{it} - \varepsilon})) \\ &= \ln(1 - 2\lambda M_{it}^* (1 - \mathbf{e}^{-\ln(\frac{M_{it}^*}{M_{it}}) + \varepsilon_{it} - \varepsilon^{(s)}})) \\ &= \ln(1 - 2\lambda \underbrace{(M_{it}^* - M_{it} \mathbf{e}^{\varepsilon_{it} - \varepsilon^{(s)}})}_{\neq e_{it}}))\end{aligned}$$

The moment to holds under ε_{it} , the realized measurement error for firm it ,
 $\ell_{it}(\varepsilon_{it}; \lambda)$

Simulating the nonlinear term

If we hold firm it 's own $(M_{it}^*, \mathcal{V}_{it})$ fixed and simulate ε ,

$$\hat{\ell}_{it}^S(\lambda) = \frac{1}{S} \sum_{s=1}^S \ell_{it}(\varepsilon^{(s)}; \lambda), \quad \varepsilon^{(s)} \stackrel{iid}{\sim} \hat{f}_{\varepsilon}$$

By the LLN, as $S \rightarrow \infty$: $\hat{\ell}_{it}^S(\lambda) \xrightarrow{a.s.} \bar{\ell}_{it}(\lambda) \equiv \int \ell_{it}(\varepsilon; \lambda) f_{\varepsilon}(\varepsilon) d\varepsilon$

In general, however,

$$\bar{\ell}_{it}(\lambda) \neq \ell_{it}(\varepsilon_{it}; \lambda)$$

Why $\mathcal{V}_{it}$ already contains $\tilde{\mathcal{W}}_{it}$

At the truth ($\varepsilon = \varepsilon_{it}$), e_{it} solves the firm's FOC: $e_{it} = e^*(\psi_{it}, \omega_{it})$ (inverting Equation 23 for e).

Substitute this and Equation 22 into Equation 21's own inverse,
 $\mathcal{V}_{it} = \ln(M_{it} + e_{it}/(M_{it})) - \varepsilon_{it}$:

$$\mathcal{V}_{it} = \ln\left(\frac{M_{it} + e^*(\psi_{it}, \tilde{\mathcal{W}}_{it} - (1 - \beta)\varepsilon_{it})}{M_{it}}\right) - \varepsilon_{it}$$

- ▶ ε_{it} must be consistent with $\mathcal{V}_{it}$ and $\tilde{\mathcal{W}}_{it}$ jointly
- ▶ ℓ_{it} must be conditioned on $(\mathcal{V}_{it}, \tilde{\mathcal{W}}_{it})$

The conditional expectation

Since $\mathcal{V}_{it}$ depends on $\tilde{\mathcal{W}}_{it}$ too (shown above), condition on both, then apply the law of iterated expectations:

$$E[\ell_{it}(\varepsilon_{it}; \lambda) \mid Z] = E\left[E[\ell_{it}(\varepsilon_{it}; \lambda) \mid Z, \mathcal{V}_{it}, \tilde{\mathcal{W}}_{it}] \mid Z\right]$$

$$E[\ell_{it}(\varepsilon_{it}; \lambda) \mid Z, \mathcal{V}_{it}, \tilde{\mathcal{W}}_{it}] = \int \ell_{it}(\varepsilon; \lambda) f_{\varepsilon|\mathcal{V}_{it}, \tilde{\mathcal{W}}_{it}, Z}(\varepsilon) d\varepsilon$$

- ▶ LIE reveals the inner integral needs $f_{\varepsilon|\mathcal{V}_{it}, \tilde{\mathcal{W}}_{it}, Z}$, the density of ε **conditional on this firm's own $\mathcal{V}_{it}$**
- ▶ $\hat{\ell}_{it}^S(\lambda)$ instead integrates against the **unconditional** f_{ε}
- ▶ Since $\mathcal{V}_{it} = u_{it} - \varepsilon_{it}$ by construction $\varepsilon_{it} \not\perp \mathcal{V}_{it}$

Testing with data

I simulate $\varepsilon^{(s)}$, $S=250$, from $\hat{f}_\varepsilon$, efficient GMM objective weighed by $\hat{\Omega}^{-1}$, and test the J-statistic against its χ^2 distribution (Hansen 1982)

$\varepsilon^{(s)}$ drawn from CORP firms' empirical distribution, feasibility-filtered ($e > 0, M > 0$);

Two set of moment conditions tested:

- ▶ Exactly identified ($d_g = d_\theta = 4$): $g(\theta) = (\psi, \psi\omega, \psi\omega^2, \varepsilon h')$ (last moment is a score term for λ). I report the value of the objective function.
- ▶ Overidentified ($d_g = 5 > d_\theta = 4$): 5th moment added, $\psi \ln M$. I report value of obj. fun and J-statistical test

Results of the test

	$\hat{L}_n (d_g = 4)$	$\hat{L}_n (d_g = 5)$	J	$\chi^2_{1,.95}$
m_{it-1}^*	1.022	1.020	29,258	3.84
$\tilde{\mathcal{W}}_{it-2}$	1.011	1.009	28,961	3.84

- ▶ Exactly identified: $\hat{L}_n$ not close to zero, parameter estimates hit generous bounds
- ▶ Overidentified: Under the null, $H_0 : g(\theta) = 0$. Rejects null at 95% by a large factor. $\hat{L}_n$ still not below 1.

Fitted Parameters

Exactly identified (4 moments)

Instrument	$\hat{\lambda}$	$\hat{\delta}_0$	$\hat{\delta}_1$	$\hat{\delta}_2$	$\hat{\sigma}_{\psi}^2$
m_{it-1}^*	1.30×10^{-4}	32.98	50.0 (box bound)	10.47	6,416
$\tilde{\mathcal{W}}_{it-2}$	1.63×10^{-8}	50.0 (box bound)	32.46	4.63	101.5

Over-identified (5 moments, adding $\psi \ln M$):

Instrument	$\hat{\lambda}$	$\hat{\delta}_0$	$\hat{\delta}_1$	$\hat{\delta}_2$	$\hat{\sigma}_\psi^2$
m_{it-1}^*	4.58×10^{-3}	-8.41	10.90	2.39	321.6
$\tilde{\mathcal{W}}_{it-2}$	1.62×10^{-5}	-50.0 (box bound)	-25.93	-3.43	81.8

- ▶ All except one specification hit a box bound on δ_0 or δ_1
- ▶ $\tilde{\mathcal{W}}_{it-2}$'s overidentified fit gets the cost function's curvature backwards ($\hat{\delta}_1, \hat{\delta}_2 < 0$)
- ▶ $\hat{\sigma}_\psi^2$ swings 82 to 6,416 across specifications
- ▶ Compare to ELVIS: $\hat{\lambda} \approx 5.4 \times 10^{-7}$ for both instruments, $\hat{\delta}_1, \hat{\delta}_2 > 0$ stable across different grid specifications

Two alternative solutions

MSL: deriving the density of observables and mapping to unobservables using a density transformation.

- ▶ Feasible but resulting density needs additional density of the AR(1) process of productivity and of the lag of ε , and integrating out two variables, ε_{it} and ε_{it-1} . MSL delivers estimates of θ and f_ψ

ELVIS: doesn't need f_ε or f_ψ or the other densities. Grid search over θ and solves for an exponential tilt of the sampler (Schennach 2014)

- ▶ Implemented in my summer paper. Strategy validated by Nail. Convolutd to explain, but practical. Preliminary estimates and counterfactual results already done.

Current fit: $\hat{\lambda} = 5.427 \times 10^{-7}$, $\hat{\delta}_1 = 4.3$, $\hat{\delta}_2 = 0.54$, $\hat{L}_n \approx 0.0002$ ELVIS

preliminary results

Maximum Simulated Likelihood (MSL)

MSL

- ▶ We could estimate f_ψ and $\theta = \{\lambda, \delta_0, \delta_1, \delta_2\}$ by Maximum Simulated Likelihood (MSL)
- ▶ After the recast, one unobservable remains: ε_{it} , with known distribution f_ε (from stage 1),
- ▶ we also know the distribution of ω
- ▶ I would need a *density transformation* to get from the primitives' density $f_{\psi, \omega, \varepsilon}(\psi, \omega, \varepsilon)$ to the density of the observables $f_{\mathcal{V}, \tilde{\mathcal{W}}}(\mathcal{V}, \tilde{\mathcal{W}})$
 - ▶ accounting for the nonlinear map from ψ to $\mathcal{V}$ (Theorem 7.5.3, next slide) *and*
 - ▶ integrating out ε <!--

$$f_{\mathcal{V}_{it}, \tilde{\mathcal{W}}_{it}}(\mathcal{V}_{it}, \tilde{\mathcal{W}}_{it}) = \int f_\psi(\psi(\varepsilon_{it})) f_\omega(\omega(\varepsilon_{it})) f_\varepsilon(\varepsilon_{it}) |\det J\Phi^{-1}(\varepsilon)| d\varepsilon_{it}$$

The object to estimate is f_ψ and $\theta = (\lambda, \delta_0, \delta_1, \delta_2)$, since f_ω and f_ε are already known →

Theorem 7.5.3 Change of Variables in R^n Judd (1998)

Assume that $\Phi : X \subset R^n \rightarrow R^n$ is C^1 and that there is a C^1 inverse map $\Phi^{-1} : Y \subset R^n \rightarrow X$ where $Y = \Phi(X)$ and Y and X are open. If f is integrable on X , then

$$\int_Y f(y)dy = \int_X f(\Phi(x)) |\det J(\Phi)(x)| dx,$$

where $J(\Phi)(x)$ is the Jacobian of Φ at x .

Corollary: Density transformation

Let $f_X(x)$ be the joint pdf of X , B any region in the support of Y , and $A = \Phi^{-1}(B)$

For $Y = \Phi(X)$: $P(X \in A) = P(\Phi(X) \in \Phi(A)) = P(Y \in B)$. Apply the theorem with $\Psi \equiv \Phi^{-1}$ in place of Φ (Φ^{-1} is C^1 with C^1 inverse Φ) to $g = f_X \cdot \mathbf{1}_A$:

$$P(X \in A) = \int_A f_X(x) dx = \int_B f_X(\Phi^{-1}(y)) |\det J\Phi^{-1}(y)| dy.$$

Since B is arbitrary, and using $J\Phi^{-1}(y) = [J\Phi(x)]^{-1}$ (inverse function theorem) at $x = \Phi^{-1}(y)$, then, the joint pdf of Y is:

$$f_Y(y) = \frac{f_X(x)}{|\det J\Phi(x)|} \Big|_{x=\Phi^{-1}(y)} \quad (26)$$

Target Density

To find the correct target density of observables, note that, given (K_{it}, L_{it}) , the variables $(\mathcal{V}_{it}, \tilde{\mathcal{W}}_{it})$ form an affine map in the (Y_{it}, M_{it}^*) -space,

$$\mathcal{V}_{it} = m_{it}^* - y_{it} + c_1 \quad (19 \text{ rearranged})$$

$$\tilde{\mathcal{W}}_{it} = y_{it} - \beta(m_{it}^* - \mathcal{V}_{it}) - \alpha_K k_{it} - \alpha_L l_{it} \quad (\text{equivalent to 20})$$

where $c_1 \equiv \ln \left(\frac{\rho_t}{P_t \beta} \right)$

Substituting the first equation into the second cancels the m_{it}^* terms exactly:

$$\tilde{\mathcal{W}}_{it} = (1 - \beta) y_{it} + \beta c_1 - \alpha_K k_{it} - \alpha_L l_{it} \quad (27)$$

Φ is a linear bijection in (y_{it}, m_{it}^*) -space

$$\Phi : (y_{it}, m_{it}^*) \mapsto (\mathcal{V}_{it}, \tilde{\mathcal{W}}_{it})$$

$$\begin{pmatrix} \mathcal{V} \\ \tilde{\mathcal{W}} \end{pmatrix} = \underbrace{\begin{pmatrix} -1 & 1 \\ 1 - \beta & 0 \end{pmatrix}}_A \begin{pmatrix} y \\ m^* \end{pmatrix} + \begin{pmatrix} c_1 \\ \beta c_1 - \alpha_K k - \alpha_L l \end{pmatrix}.$$

Since $\det A = -(1 - \beta) \neq 0$ ($0 < \beta < 1$) $\implies \Phi$ is an affine bijection

- ▶ $\implies (\mathcal{V}_{it}, \tilde{\mathcal{W}}_{it})$ contain the same information as (Y_{it}, M_{it}^*)
- ▶ Therefore, the correct target density $f_{\mathcal{V}, \tilde{\mathcal{W}}|K, L}$
- ▶ **No need** to include M^* in the joint density or condition on $(\mathcal{V}$ contains all information of $M^*)$

Conditioning on ω_{it-1} instead of (K_{it}, L_{it})

Due to the Markov process of productivity, $\omega_{it} = g(\omega_{it-1}) + \eta_{it}$,
 $\eta_{it} \perp\!\!\!\perp (\omega_{it-1}, k_{it}, l_{it})$,

$$f_{\omega_{it}|\omega_{it-1}, K_{it}, L_{it}}(\omega_{it}) = f_{\omega_{it}|\omega_{it-1}}(\omega_{it}) = f_{\eta}(\omega_{it} - g(\omega_{it-1}))$$

Conditional on ω_{it-1} , the only randomness left in ω_{it} is η_{it} , which doesn't depend on (K_{it}, L_{it})

► ω_{it-1} is a **sufficient statistic**: once it's known, (K_{it}, L_{it}) adds nothing.

Setting Φ in the $(\psi_{it}, \omega_{it}, \varepsilon_{it})$ -space

$\Phi : (\psi_{it}, \omega_{it}, \varepsilon_{it}) \mapsto (\mathcal{V}_{it}, \tilde{\mathcal{W}}_{it}, a)$, padding with the auxiliary $a \equiv \varepsilon_{it}$ so
 $\Phi : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ matches the theorem's dimension requirement:

Define $G(\psi_{it}, \omega_{it}) \equiv u(e^*(\psi_{it}, \omega_{it}))$,

- ▶ where $u(e) = \ln((M_{it} + e)/M_{it})$, holding M_{it} fixed as e varies $(M \perp e \mid \omega)$, and
- ▶ $e^*(\psi_{it}, \omega_{it}) = h^{-1}(\delta_0 - \delta_1 \omega_{it} + \delta_2 \omega_{it}^2 + \psi_{it})$ inverts the FOC for e_{it} ,

$$\mathcal{V} = G(\psi, \omega) - \varepsilon, \quad \tilde{\mathcal{W}} = \omega + (1 - \beta)\varepsilon, \quad a = \varepsilon$$

- ▶ Like (K_{it}, L_{it}) before, ω_{it-1} does not enter the map

Conditional density

Let $J_i(\varepsilon_{it}) = |\det J\Phi^{-1}|$, then,

$$f_{\mathcal{V}_{it}, \tilde{\mathcal{W}}_{it} | \omega_{it-1}}(\mathcal{V}_{it}, \tilde{\mathcal{W}}_{it}) = \int f_{\psi}(\psi(\varepsilon_{it})) J_i(\varepsilon_{it}) f_{\eta}(\omega(\varepsilon_{it}) - g(\omega_{it-1})) f_{\varepsilon}(\varepsilon_{it}) d\varepsilon_{it}. \quad (28)$$

Assuming $\psi_{it}, \varepsilon_{it} \perp (K_{it}, L_{it}) \mid \omega_{it-1}$

Since ω_{it-1} isn't observed, use $\tilde{\mathcal{W}}_{it-1}$

$\tilde{\mathcal{W}}_{it-1}$ is firm- i *data*

- ▶ ε_{it-1} unobserved, but known density f_ε (stationarity) and
- ▶ already assumed independent $\varepsilon_{it-1} \perp \varepsilon_{it}, \eta_{it}$ in productivity's AR(1) estimation
- ▶ extending $\varepsilon_{it-1} \perp \psi_{it}$ (additional assumption)
- ▶ therefore, we can marginalize ε_{it-1} with an ordinary integral,

Likelihood contribution of a firm

$$\begin{aligned} f_{\mathcal{V}_{it}, \tilde{\mathcal{W}}_{it} | \tilde{\mathcal{W}}_{it-1}}(\mathcal{V}_{it}, \tilde{\mathcal{W}}_{it} | \tilde{\mathcal{W}}_{it-1}) = \\ \iint f_{\psi}(\psi(\varepsilon_{it})) J_i(\varepsilon_{it}) f_{\eta}\left(\omega(\varepsilon_{it}) - g(\tilde{\mathcal{W}}_{it-1} - (1 - \beta)\varepsilon_{it-1})\right) \\ \times f_{\varepsilon}(\varepsilon_{it}) f_{\varepsilon}(\varepsilon_{it-1}) d\varepsilon_{it} d\varepsilon_{it-1}. \end{aligned} \tag{29}$$

Recovering f_η

Define the residual used in the AR(1) productivity process,

$$\begin{aligned}\zeta_{it} &\equiv \mathcal{W}_{it} - \alpha_K k_{it} - \alpha_L l_{it} - \gamma_0 - \gamma_1 (\mathcal{W}_{it-1} - \alpha_K k_{it-1} - \alpha_L l_{it-1}) \\ &= (1 - \beta) \varepsilon_{it} - \gamma_1 (1 - \beta) \varepsilon_{it-1} + \eta_{it},\end{aligned}$$

a sum of three mutually independent variables, two with known densities, f_ε .
Fourier deconvolution recovers the third:

$$\phi_\eta(t) = \frac{\phi_\zeta(t)}{\phi_\varepsilon((1 - \beta)t) \phi_\varepsilon(-\gamma_1(1 - \beta)t)} \longrightarrow f_\eta,$$

The Jacobian matrix

$$J\Phi = \frac{\partial(\mathcal{V}, \tilde{\mathcal{W}}, a)}{\partial(\psi, \omega, \varepsilon)} = \begin{pmatrix} G_\psi & G_\omega & -1 \\ 0 & 1 & 1 - \beta \\ 0 & 0 & 1 \end{pmatrix}$$

where $G_\psi \equiv \partial G / \partial \psi$ and $G_\omega \equiv \partial G / \partial \omega$

Determinant

Expand along column 1 (only G_ψ is nonzero there):

$$\det J\Phi = G_\psi \cdot \det \begin{pmatrix} 1 & 1 & -\beta \\ 0 & 1 & 1 \end{pmatrix} = G_\psi \cdot 1 = G_\psi$$

- ▶ G_ω never enters because every cofactor multiplying it comes from a zero in column 1.
- ▶ Mechanically: $\tilde{\mathcal{W}} = \omega + (1 - \beta)\varepsilon$ has **unit coefficient on ω** , a flat translation, so it contributes no curvature.
- ▶ A Jacobian only appears where the map is genuinely curved: the $\psi \rightarrow \mathcal{V}$ link through G , not $\omega \rightarrow \tilde{\mathcal{W}}$.

Unpacking G_ψ

$G = u \circ e^*$, so by the chain rule $G_\psi = u'(e^*) \cdot e_\psi^*$.

► $u'(e)$: from $u(e) = \ln(M_{it} + e) - \ln M_{it}$, with M_{it} fixed,

$$u'(e) = \frac{1}{M_{it} + e}$$

evaluated at $e = e^*$, $M_{it} + e^* = M_{it}^*$, $\implies u'(e) = 1/M_{it}^*$

► e_ψ^* : differentiate the FOC identity $\psi = h(e^*(\psi, \omega)) - \delta_0 + \delta_1\omega - \delta_2\omega^2$ w.r.t. ψ (implicit function theorem, valid since $h' \neq 0$):

$$1 = h'(e^*) \cdot e_\psi^* \implies e_\psi^* = \frac{1}{h'(e^*)}$$

Combine:

$$G_\psi = \frac{1}{M_{it}^* h'(e)}$$

Assembling $J_i(\varepsilon)$

From Equation 26, the multiplying factor is $1/|\det J\Phi| = 1/|G_\psi|$:

$$J_i(\varepsilon) \equiv \frac{1}{|G_\psi|} = |h'(e)| M_{it}^*$$

Every piece is traceable: - $|h'(e)|$ from inverting the FOC ($\psi \rightarrow e$), - M_{it}^* from the log-ratio definition ($e \rightarrow \mathcal{V}$, holding the *true* M_{it} fixed) - they multiply because $G = u \circ e^*$ is a **composition**, so the chain rule stacks them.

Marginalizing out the auxiliary

Applying the density formula gives the joint density of $(\mathcal{V}, \tilde{\mathcal{W}}, a)$, but only $(\mathcal{V}, \tilde{\mathcal{W}})$ are observed;

- ▶ Integrating a out (an ordinary single integral, not part of the theorem) gives the likelihood contribution of firm i :

$$f_{\mathcal{V}, \tilde{\mathcal{W}}}(\mathcal{V}_{it}, \tilde{\mathcal{W}}_{it}) = \int f_{\psi}(\psi(\varepsilon)) J_i(\varepsilon) f_{\omega}(\omega(\varepsilon)) f_{\varepsilon}(\varepsilon) d\varepsilon \quad (30)$$

A 3-for-3 change of variables that ends in a 1-D integral:

- ▶ the theorem handles the square transformation and hands back a Jacobian;
- ▶ marginalizing the unobserved auxiliary coordinate is the separate, ordinary step left over once the theorem's job is done.

Simulated log-likelihood

$$\mathcal{L}(\theta, f_\psi) = \sum_{i \in \text{Unincorp}} \ln \left(\int f_\psi(\psi(\varepsilon)) J_i(\varepsilon) f_\omega(\omega(\varepsilon)) f_\varepsilon(\varepsilon) d\varepsilon \right) \quad (31)$$

Simulating the likelihood

Approximate Equation 30 by **Monte Carlo**

$$\widehat{\mathcal{L}}_i(\theta) = \frac{1}{S} \sum_{s=1}^S f_{\psi}(\psi(\varepsilon^{(s)})) J_i(\varepsilon^{(s)}) f_{\omega}(\omega(\varepsilon^{(s)})), \quad \varepsilon^{(s)} \stackrel{iid}{\sim} \hat{f}_{\varepsilon} \quad (32)$$

- ▶ Draws $\varepsilon^{(s)}$ come from *exactly* the density $\hat{f}_{\varepsilon}(\varepsilon) d\varepsilon$ multiplying the integrand — this is plain Monte Carlo integration, unbiased for **any** $S \geq 1$
- ▶ Alternatively, use **deterministic quadrature** to integrate Equation 30

Why this works despite ε entering e nonlinearly

- ▶ $e(\varepsilon) = M_{it}^* (1 - \mathbf{e}^{-\mathcal{V}_{it} - \varepsilon})$ is nonlinear in ε
- ▶ But Equation 32 averages the **density value** $f_{\psi}(\psi(\varepsilon^{(s)})) J_i(\varepsilon^{(s)}) f_{\omega}(\omega(\varepsilon^{(s)}))$ at each draw
 - ▶ never averages $\psi(\varepsilon^{(s)})$ *by itself*
- ▶ Monte Carlo integration of a (possibly wild) function against its own sampling density is unbiased regardless of the function's shape
 - ▶ nonlinearity only breaks things if the reweighting is skipped
- ▶ Every simulated $\varepsilon^{(s)} \neq$ the firm's true ε_{it} , almost surely
 - ▶ but draws implying an implausible (ψ, ω) pair get down-weighted by $f_{\psi} \cdot J_i \cdot f_{\omega}$; only draws near the truth carry weight,
 - ▶ so the weighted average reconstructs the correct posterior-weighted integral even though no single draw is the truth

Convergence — a known theorem

Lee (1992): if the simulator of a likelihood contribution is (i) unbiased and continuous in θ , and (ii) built from draws i.i.d. across firms and replicates, then under standard M-estimation regularity (compactness, identification, dominance) the simulated-MLE is consistent and $\sqrt{N}$ -equivalent to the infeasible exact MLE, **provided** $\sqrt{N}/S \rightarrow 0$ as $N, S \rightarrow \infty$ (fixed S leaves an $O(1/S)$ bias, since $\ln(\text{average}) \neq \text{average of } \ln$)

- ▶ Unbiasedness (i): shown above, holds by construction
- ▶ Continuity (ii): h , q , and $J_i(\varepsilon)$ are smooth throughout the interior domain $e \in (0, \frac{1}{2\lambda})$
- ▶ Draws i.i.d. across i, t, s , independent of θ : by construction

An even cleaner argument: quadrature

- ▶ The preferred implementation is **deterministic quadrature**, not Monte Carlo simulation, for this 1-D integral
- ▶ Quadrature avoids simulation noise because it is deterministic and can be driven arbitrarily small by adding nodes
- ▶ The $\sqrt{N}/S \rightarrow 0$ condition above is then moot: with negligible quadrature error, ordinary MLE consistency applies directly

Caveat: this is a two-step estimator

- ▶ β and f_ε are **first-stage estimates**, not the truth
- ▶ Still consistent under standard conditions, but naive stage-2 standard errors ignore stage-1 estimation uncertainty
- ▶ Use the Murphy and Topel (1985) two-step correction, or bootstrap both stages jointly

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